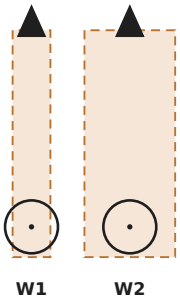
T4 Book interconnects

Print sheets 1 and 2 on the two sides of one card, flipping on the long edge. Make three.

V2 bench test



1 Edge wraps

Lay each W lane on side A, run the tape over the edge and down side B to the test point behind. W1 and W2 cross the top edge, W3 and W4 the outer edge, W5 and W6 the bottom edge. W1, W3 and W5 are 5 mm tape; W2, W4 and W6 are 12 mm.

2 Links at the edge and inboard

E1 At the edge
solder bridge across the wrapped edges

H1 · no net

3 The spine contract

15 mm pitch · 5 mm tape · 10 mm clear gap

C1 GND

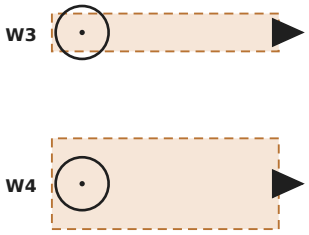
C2 BUS A

C3 V+

ring axis

C4 BUS B

C5 GND



H2 · no net

E2 At the edge
stranded wire, 30 mm service loop

E3 At the edge
one ribbon conductor, 30 mm loop

I1 10 mm inboard
stranded wire, 30 mm loop

I2 10 mm inboard
one ribbon conductor, 30 mm loop

X1 On the axis
stranded wire laid 15 mm along it

X2 On the axis
one ribbon conductor laid along it

X3 On the axis
wire under a stitched card patch

X4 On the axis
solder bridge

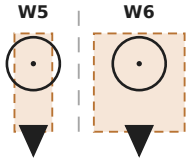
H3 · no net

X5 Your version

4 Links on the ring axis

Print three cards: pages 1, 2 and 3 of a test book. Side A, this side, takes each link from the page before; side B sends it on to the page after.

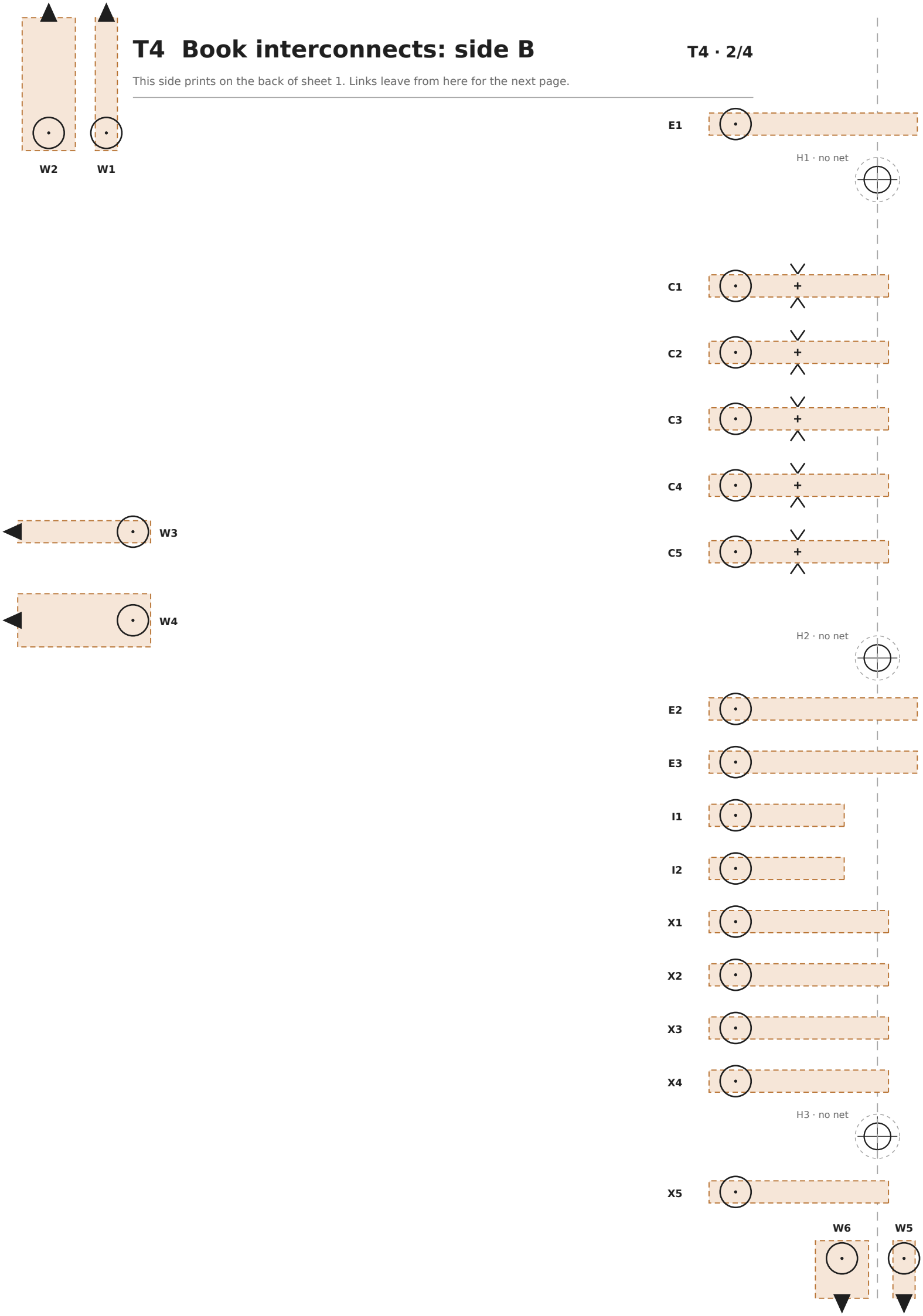
With the book open, the back of one page lies beside the front of the next, so both ends of a link can be probed at once. Sheet 3 says how to build and test the links.



Print at 100 %: this bar should measure 100 mm.

T4 Book interconnects: side B

This side prints on the back of sheet 1. Links leave from here for the next page.



T4 Running the book interconnect tests

Print sheets 3 and 4 once on plain paper. Print sheets 1 and 2 double-sided on card, three times.


H1 · no net

You need

- Three cards printed double-sided (sheets 1 and 2), a three-hole punch and a ring binder
- 5 mm and 12 mm copper tape, solder and a soldering iron
- Stranded hookup wire, about 28 AWG, and a strip of ribbon cable
- Thin card, clear tape, a needle and thread for strain relief
- A 100 Ω resistor (½ W or more), a 3 × AA holder and clip leads
- A multimeter with a 200 mV range

1 · Wraps

1. Lay W1 to W6 on side A of one card, run the tape over the edge and on down side B to the test point behind. Burnish the fold.

2. Measure each wrap from its side A test point to its side B test point, as in step 3.

3. Bend each wrapped edge back and forth ten times and measure again. Look for cracks.

2 · Build the test book

1. On all three cards, lay every link lane (E1 to X5) and the contract lanes (C1 to C5) on both sides. Keep all five contract lanes separate: do not join C1 to C5 at the contract block. Keep the clear gap between adjacent lanes at least twice as wide as the tape itself.

2. Make the contract block's wire vias: push a cut-off resistor leg through each via mark and solder it on both sides. Each via is an independent route between sides A and B. The inward chevrons identify the through-page construction; round circle-and-dot marks are electrical test points. At each solid arrow, continue the tape to the physical edge and fold it.


H2 · no net

3. Bind the cards in order, page 1 on top.

4. For every row, join the back of page 1 to the front of page 2, and the back of page 2 to the front of page 3, the way the row's label says. Keep each solder joint at least 5 mm from the ring axis, except X4's, which tests a joint on it. On the axis rows, lay the wire or ribbon along the axis between the two joints.

5. Join each of C1 to C5 to its matching pin on the next page with a five-conductor strip of ribbon laid along the axis. Do not bridge C1 and C5 in the contract footprint. Mark the C1 edge with a stripe and keep that stripe on C1 at both ends; reversing the ribbon swaps BUS A and BUS B even though V+ and GND stay safe.

3 · Measuring a link

1. Clip the holder, the 100 Ω resistor and the link in a loop: holder +, resistor, the link's test point on the back of one page, the link, its test point on the front of the next page, holder −.

2. Measure the voltage across the resistor and divide by 100 Ω: that's the current, about 45 mA.

3. On the 200 mV range, measure from one test point to the other. 4.5 mV at 45 mA is 0.1 Ω.

4 · The spine contract chain

1. Temporarily join C1 and C5 with clip leads on page 1, outside the contract footprint. Clip the holder to page 1: red to C3 (V+), black to that temporary GND junction.

2. Temporarily join C1 and C5 on page 3. Clip the 100 Ω resistor across C3 and that temporary GND junction.

3. Measure C3 from one page to the next. Its drop divided by the total current must be 0.1 Ω or less.

4. Measure between the temporary GND junctions from one page to the next. The drop divided by total current must be 0.05 Ω or less. Do not calculate either GND conductor from total current; the current splits.

5. Remove the temporary GND joins. Test C1, C2, C4 and C5 one at a time in the separate loop from section 3; each must be 0.1 Ω or less. Confirm that the ribbon's striped edge is on C1 at both ends before applying power.

5 · Flex cycles

1. One cycle: open the spread between pages 1 and 2 all the way, close it, then do the same for pages 2 and 3.

2. Measure every link after 10, 100, 500 and 1,000 cycles.

3. While turning the pages slowly, watch the meter across the load on page 3. Any jump is a flicker; note which link moved.

4. Check every joint with a magnifier for cracks, lifted tape and torn paper.

6 · Repair

1. Unclip the holder. Cut every link to page 2, take it out, bind a fresh page 2 and resolder.

2. Note how long it took, then measure page 2's links again.

Pass or fail

- **Pass:** each individual link and contract conductor is 0.1 Ω or less, the temporarily paired GND conductors are 0.05 Ω or less, and the load stays steady while the pages turn.
- **Fail:** a crack, lifted tape, torn paper or any flicker.
- Links at the edge move about 24 mm each cycle. If they fail, the V2 standard will prohibit them.


H3 · no net

T4 Results

Readings in mV at the measured current. Divide by the current for ohms.

Wraps, side A to side B

	New	Bent	Cracks?
W1			
W2			
W3			
W4			
W5			
W6			

The spine contract chain

Current, from the load: _____ mA

	1 to 2	2 to 3
C1 GND 1		
C2 BUS A		
C3 V+		
C4 BUS B		
C5 GND 2		
C1 + C5 GND pair		

Results for the standard

Best link construction	
Pin-1 stripe stayed on C1 at every joint	
Closest safe distance from the axis	
Service loop needed	
Wrap tape: 5 mm or 12 mm	
Flex-life target met (cycles)	
Constructions to prohibit	

Links, page 1 to page 2

	0	10	100	500	1,000	Flicker?
E1						
E2						
E3						
I1						
I2						
X1						
X2						
X3						
X4						
X5						

Links, page 2 to page 3

	0	10	100	500	1,000	Flicker?
E1						
E2						
E3						
I1						
I2						
X1						
X2						
X3						
X4						
X5						